CSE 598 — Spring 2026

τ-Bench Phase 2: Error Analysis Report

τ-Bench: Language Agent Evaluation

Group Members

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1. Introduction

This report presents a systematic error analysis of tool-calling language agents evaluated on the τ-bench benchmark. In Phase 1, we established baseline performance using three strategies (ReAct, ACT, Function Calling) across Qwen3 model sizes (4B, 8B, 14B, 32B) on both airline and retail domains. The overall success rate across 6,282 trajectories was 19%, confirming fundamental reliability issues across all configurations.

Phase 2 diagnoses why these baselines fail. We developed a custom error taxonomy with 8 error types organized into a 5-level classification hierarchy. We then applied automated classification to all 6,282 trajectories (5,090 failures), revealing distinct error profiles that shift dramatically with model size — a finding that directly informs our multi-agent system proposal.

Our taxonomy is independently developed and differs from both the τ-bench paper's 4-category taxonomy and the IRMA paper's taxonomy. Our categories are automatable — each error type is detected programmatically by comparing agent trajectories against ground truth, enabling large-scale quantitative analysis.

2. Error Taxonomy

2.1 Four-Level Classification Hierarchy

Our classification follows a hierarchical decision tree that progressively narrows the set of possible failure diagnoses. Each trajectory is checked at increasingly specific levels:

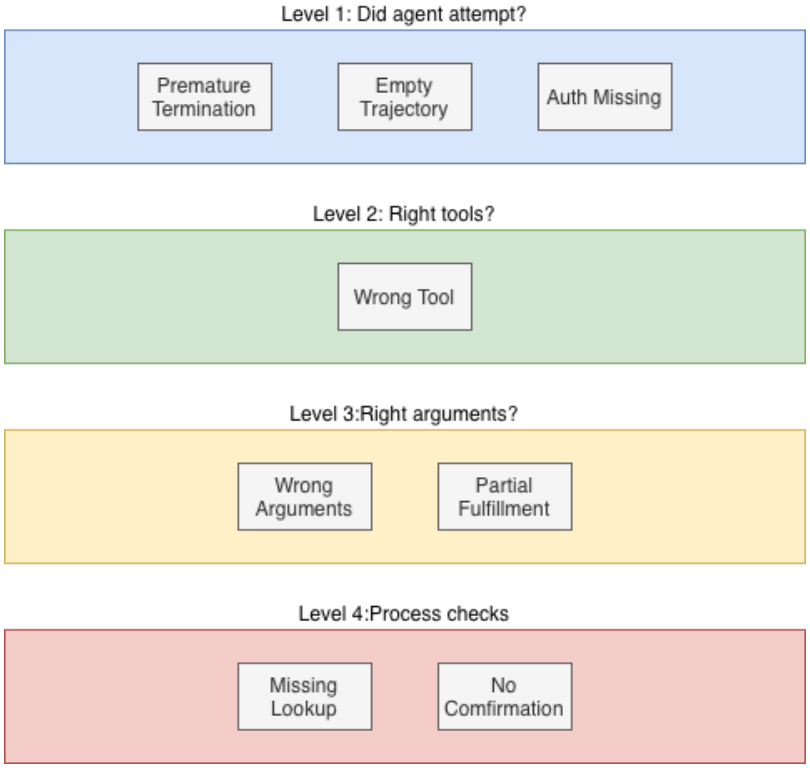

Figure 1: Four-level error classification hierarchy with aggregate counts across all models (6,282 trajectories).

Level 1 — Did the agent attempt the task?

PREMATURE_TERMINATION (1,626 total) — Agent never called the required consequential (write) tool. Includes hallucinated completions.

EMPTY_TRAJECTORY (1,325 total) — Server crash or timeout — no trajectory data recorded.

AUTH_MISSING (782 total) — Agent skipped user authentication, often cascading into fabricated data.

Level 2 — Correct tools?

WRONG_TOOL (1,327 total) — Agent called a different consequential tool than required.

Level 3 — Correct arguments?

WRONG_ARGUMENT (997 total) — Correct tool called with incorrect arguments. This is a combined category encompassing 6 sub-types: wrong order ID (253), wrong item selection (231), wrong replacement item (221), wrong payment method (118), other argument mismatches (162), and wrong reason/justification (12).

PARTIAL_FULFILLMENT (40 total) — Correct tool and correct arguments, but incomplete item coverage (e.g., returned 2 of 3 requested items).

Level 4 — Proper process?

MISSING_LOOKUP (1,045 total) — Fewer read/lookup tool calls than ground truth expected.

NO_CONFIRMATION (626 total) — Executed consequential tool without obtaining user confirmation.

2.2 Cause-Oriented Grouping

For actionability, we group the 8 types into 6 cause-oriented categories. Each maps to a potential multi-agent intervention:

Table 1: Cause-Oriented Error Categories (aggregate across all models)

Category	Error Types	Count	Description
Execution Failure	Premature Termination, Empty Trajectory	2,951	The agent never executes a consequential action
Tool Selection Error	Wrong Tool	1,327	Fundamentally different operation chosen
Argument Error	Wrong Argument	997	Correct tool, wrong entity identifiers or parameters
Authentication Bypass	Auth Missing	782	Agent skips user identification step
Incomplete Resolution	Partial Fulfillment, Missing Lookup	1,085	Task partially completed or info skipped
Protocol Violation	No Confirmation	626	Right action, violated process rules

2.3 Comparison with τ-bench Paper Taxonomy

The τ-bench paper defines 4 categories from manual analysis of 36 GPT-4o FC trajectories: wrong argument (33%), wrong decision (25%), wrong info (22%), partially resolved (19%). Our taxonomy differs in three key ways:

- **Granularity:** 8 types vs. 4 — their 'wrong argument' maps to our WRONG_ARGUMENT category, but we additionally capture sub-type distributions.
- **Automation:** Our categories are defined by programmatic comparison rules, enabling analysis of 4,937 trajectories (vs. 153 manual annotations).
- **Coverage:** We capture PREMATURE_TERMINATION (hallucinated completion), AUTH_MISSING, and NO_CONFIRMATION — absent from their taxonomy but accounting for 59.6% of our failures.

3. Quantitative Analysis

3.1 Dataset Overview

We analyzed 6,282 trajectories across 19 configurations (3 models × 2 domains × 3 baselines), yielding 1,192 successes (18.97%) and 5,090 failures (81.02%). The 8B model data was unavailable due to infrastructure issues.

Table 2: Per-Configuration Breakdown (19 groups)

Model	Domain	Baseline	Total	Pass	Fail	Rate
4B	Airline	Act	250	81	169	32.4%
4B	Airline	ReAct	250	88	162	35.2%
4B	Airline	FC	250	53	197	21.2%
4B	Retail	Act	570	50	520	8.8%
4B	Retail	ReAct	570	66	504	11.6%
4B	Retail	FC	570	51	519	8.9%
8B	Retail	FC	88	13	75	14.8%
14B	Airline	Act	250	61	189	24.4%
14B	Airline	ReAct	250	61	189	24.4%
14B	Airline	FC	250	48	202	19.2%
14B	Retail	Act	490	69	421	14.1%
14B	Retail	ReAct	495	105	390	21.2%
14B	Retail	FC	511	96	415	18.8%
32B	Airline	Act	191	43	148	22.5%
32B	Airline	ReAct	75	19	56	25.3%
32B	Airline	FC	157	39	118	24.8%
32B	Retail	Act	396	81	315	20.5%
32B	Retail	ReAct	222	74	148	33.3%
32B	Retail	FC	447	94	353	21.0%

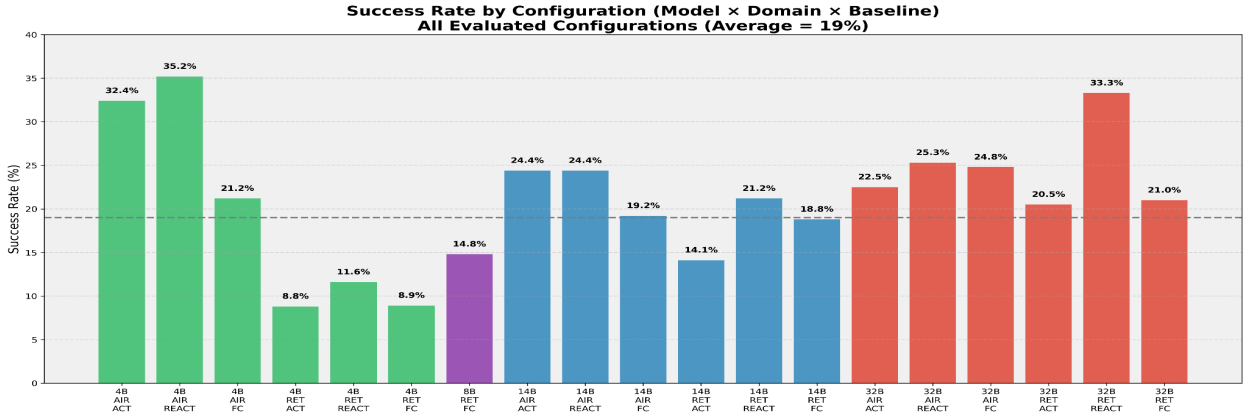

Figure 2: Success rate across all 19 configurations. Dashed line = overall average (19%).

3.2 Phase 1 Pass^k Summary

Table 3: Phase 1 Pass^k Results

Model	Domain	Baseline	P^1	P^2	P^3	P^4	P^5
32B	Airline	Act	0.172	0.040	0.006	0.000	0.000
32B	Airline	ReAct	0.076	0.010	0.000	0.000	0.000
32B	Airline	FC	0.156	0.040	0.008	0.000	0.000
32B	Retail	Act	0.141	0.047	0.022	0.012	0.009
32B	Retail	ReAct	0.130	0.019	0.003	0.000	0.000
32B	Retail	FC	0.165	0.059	0.031	0.018	0.009
14B	Airline	Act	0.244	0.162	0.132	0.112	0.100
14B	Airline	ReAct	0.244	0.166	0.126	0.100	0.080
14B	Airline	FC	0.192	0.074	0.034	0.020	0.020
14B	Retail	Act	0.141	0.033	0.012	0.004	0.000
14B	Retail	ReAct	0.212	0.065	0.028	0.015	0.009
14B	Retail	FC	0.188	0.046	0.014	0.004	0.000
4B	Airline	Act	0.324	0.254	0.218	0.196	0.180
4B	Airline	ReAct	0.352	0.312	0.294	0.284	0.280
4B	Airline	FC	0.212	0.114	0.068	0.040	0.020
4B	Retail	Act	0.087	0.034	0.019	0.012	0.008
4B	Retail	ReAct	0.115	0.0623	0.043	0.033	0.026
4B	Retail	FC	0.089	0.036	0.018	0.012	0.008

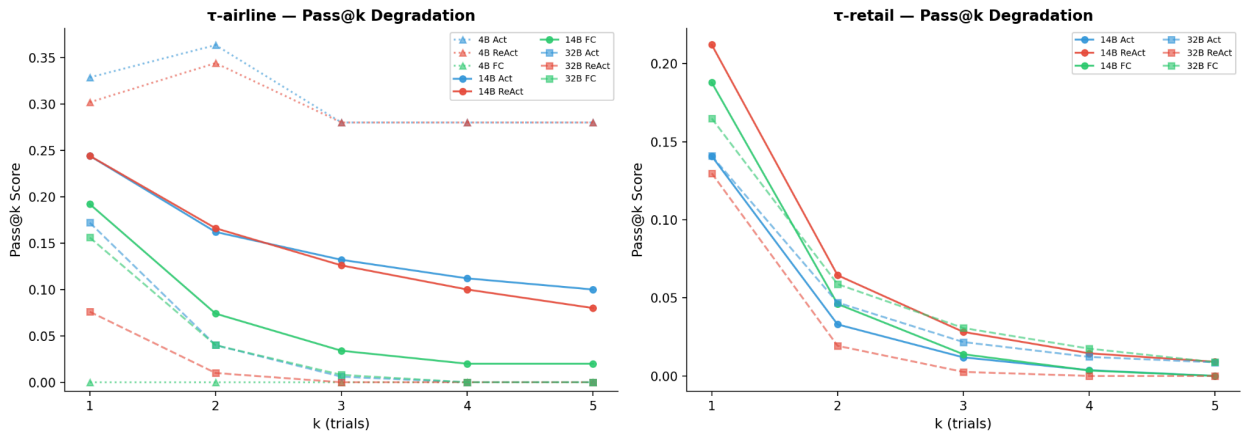

Figure 3: Pass^k degradation curves across models. Solid=14B, dashed=32B, dotted=4B.

3.3 Aggregate Error Distribution

Across all 5,090 failures, PREMATURE_TERMINATION (31.9%), WRONG_TOOL (26.1%), and EMPTY_TRAJECTORY (26.0%) are the most frequent error types. EMPTY_TRAJECTORY represents infrastructure failures (server crashes, timeouts) rather than agent behavior, so we focus our analysis on the remaining error types. Note that a single trajectory can trigger multiple error types simultaneously.

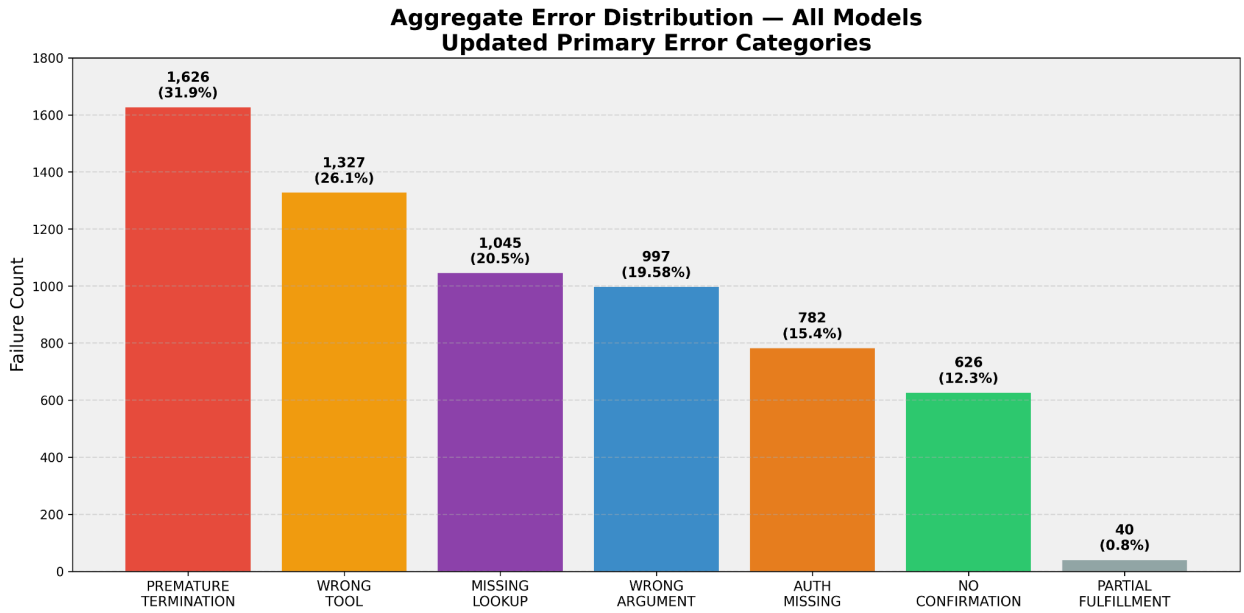

Figure 4: Aggregate error distribution across all models (excluding EMPTY_TRAJECTORY).

Table 4: Complete Error Distribution — All Models (5,090 failures)

Error Type	Count	% Failures	Cause Group
PREMATURE_TERMINATION	1,626	31.9%	Execution Failure
WRONG_TOOL	1,327	26.1%	Tool Selection Error

EMPTY_TRAJECTORY	1,325	26.0%	Infrastructure
MISSING_LOOKUP	1,045	20.5%	Incomplete Resolution
WRONG_ARGUMENT	997	19.58%	Argument Error
AUTH_MISSING	782	15.4%	Authentication Bypass
NO_CONFIRMATION	626	12.3%	Protocol Violation
PARTIAL_FULFILLMENT	40	0.8%	Incomplete Resolution

3.4 Cross-Model Error Comparison

The error distribution varies dramatically by model size. Figure 5 shows the normalized error rates (% of each model's failures) side-by-side:

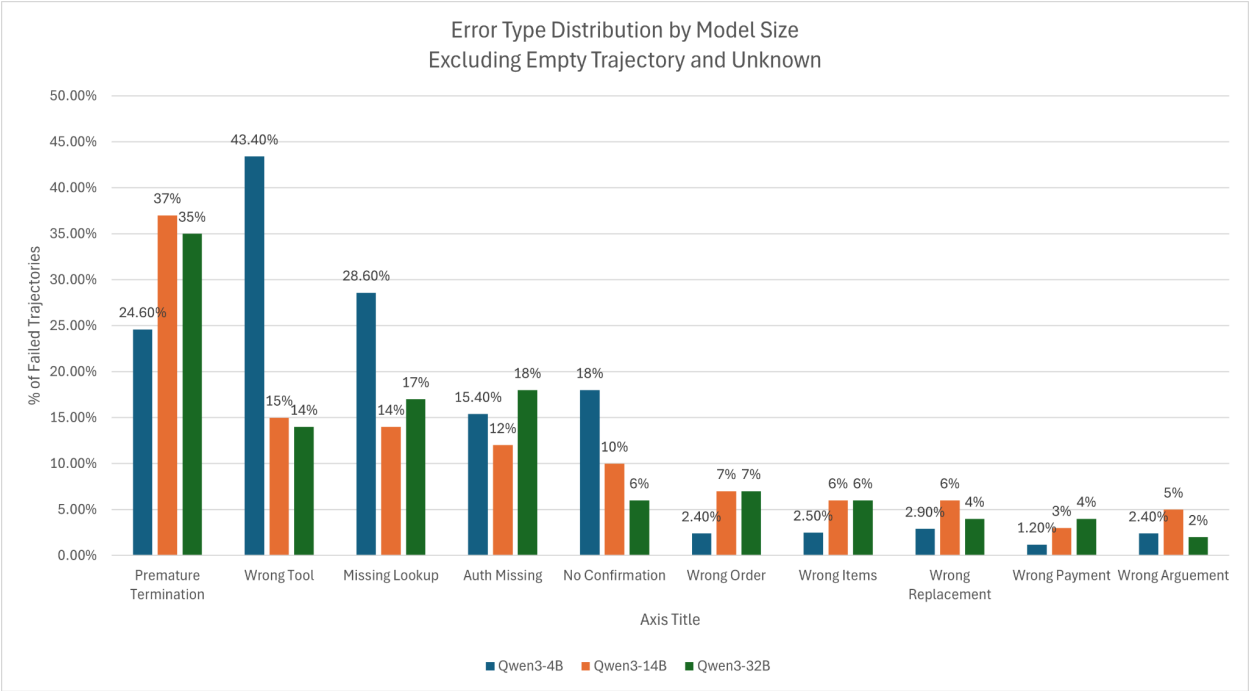

Figure 5: Error type distribution by model size (normalized %). 4B is dominated by WRONG_TOOL; 14B/32B by PREMATURE_TERMINATION.

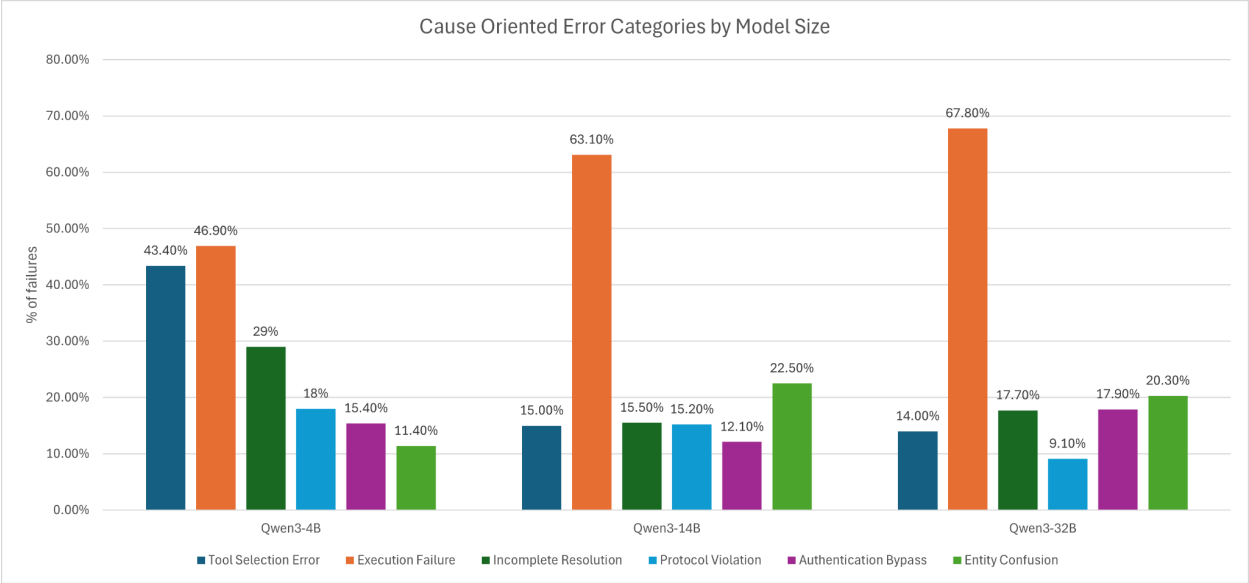

Figure 6: Cause-oriented categories by model, showing how failure profiles shift with scale.

3.5 Error Profile Shift by Model Size

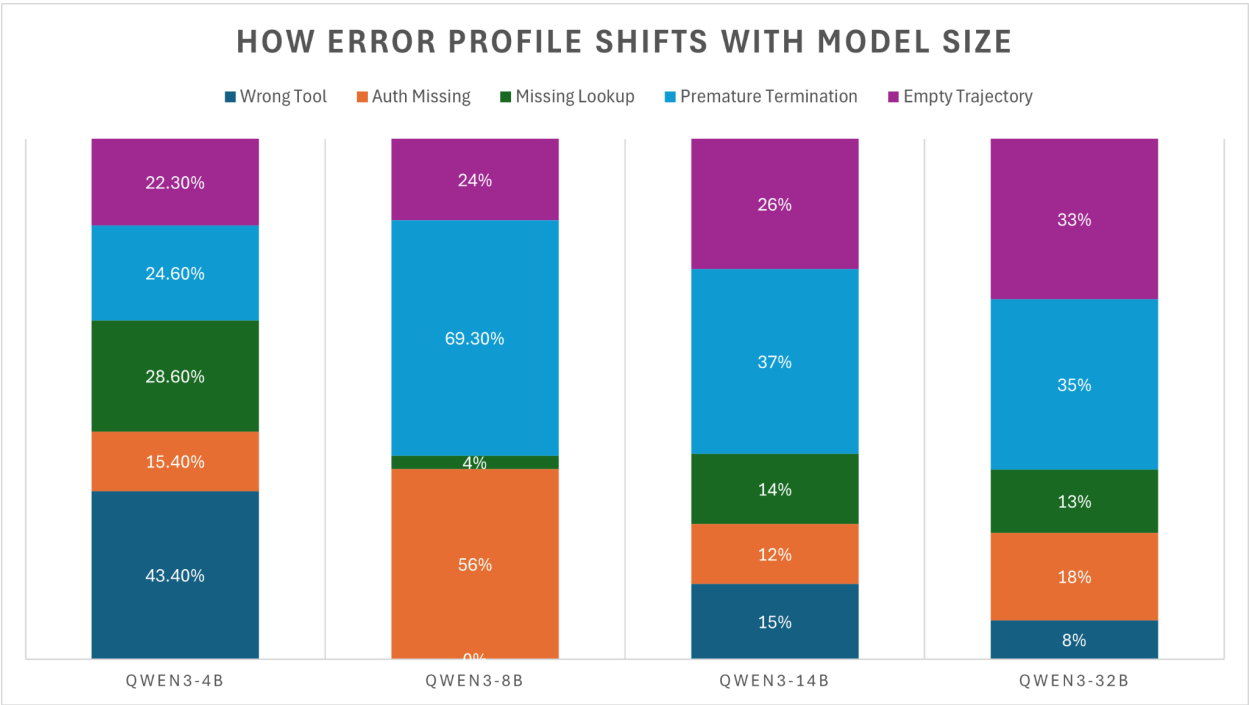

Figure 7: Stacked error profiles showing the dominant failure mode shifts from Tool Selection (4B) to Execution Failure (8B/14B/32B).

Figure 7 reveals the most important cross-model finding: as model size increases, the dominant failure mode shifts fundamentally.

Qwen3-4B: WRONG_TOOL dominates at 43.4% — the model has sufficient instruction-following capacity to attempt tool calls but lacks the reasoning to select the correct tool. PREMATURE_TERMINATION is at 24.6%, meaning the 4B model at least tries.

Qwen3-8B: PREMATURE_TERMINATION is at a high of 69.3% and is close to AUTH_MISSING, which is at 56%.

Qwen3-14B: PREMATURE_TERMINATION jumps to 37.0% — the model is now capable enough to generate plausible completion text, but this same fluency enables hallucinated completions. WRONG_TOOL drops to 15.0% as tool selection reasoning improves.

Qwen3-32B: PREMATURE_TERMINATION remains high at 34.8%, and AUTH_MISSING rises to 17.9% (vs. 12.1% in 14B). The largest model is most prone to confidently skipping required steps and fabricating entire interaction sequences.

Table 5: Per-Model Summary

Model	Total	Successes	Failures	Top Error
4B	2,460	389 (15.8%)	2,071	WRONG_TOOL (43.4%)
8B	75	13 (14.8%)	75	PREMATURE_TERM (69.3%)
14B	2,246	440 (19.6%)	1,806	PREMATURE_TERM (37.0%)
32B	1,488	350 (23.5%)	1,138	PREMATURE_TERM (34.8%)
All	**6,282**	**1,192 (19%)**	**5,090**	**PREMATURE_TERM (31.9%)**

3.6 Pass^k Degradation Analysis

The pass^k degradation curves (Figure 3) show that 14B achieves the best consistency. On airlines, the 14B Act maintains pass^5 = 0.10, while the 32B drops to zero for all baselines. The 4B model shows artificially stable curves on airline Act/ReAct due to the limited task set. Function Calling shows the steepest degradation across all models, suggesting structured tool-calling formats are more brittle despite sometimes achieving competitive pass^1.

4. Representative Failure Analysis

This section presents representative failure patterns from manual review of sampled trajectories. Full trajectory JSON files (5+ per error category) are provided in the accompanying submission.

4.1 Premature Termination (1,626 occurrences)

The most impactful error type. From manual review of sampled 32B trajectories, we identified three sub-patterns:

Sub-pattern B: Hallucinated Completion (~70%) — The agent generates plausible confirmation text like 'Your exchange has been processed! Your refund of $109.27 will appear in 5–7 business days' — with ZERO tool calls. The agent fabricates order details, tracking numbers, and refund amounts. This is the most dangerous pattern because it appears successful to an unsupervised system.

Sub-pattern A: Never Started (~20%) — The agent gets stuck asking for information it already has or loops without progressing.

Sub-pattern C: Derailed Mid-Task (~10%) — The agent begins correctly but gets confused mid-task and runs out of turns.

4.2 Tool Selection Errors — 4B Model (240 occurrences, 53.7%)

The 4B model's dominant failure is calling the wrong tool entirely. Unlike larger models, which hallucinate completions, 4B at least attempts tool calls — it just picks the wrong ones. Common patterns include calling cancel_pending_order when exchange_delivered_order_items is required, or calling modify_pending_order_items for a return request. This suggests the 4B model can follow the tool-calling format but lacks the reasoning to match user intents to appropriate tool APIs.

4.3 Argument Errors & Protocol Violations

WRONG_ARGUMENT (997 total) represents cases where the agent selects the correct tool but passes incorrect parameters. The 6 sub-types within this category reveal specific failure patterns: wrong order ID (253 cases) often from operating on the wrong order when users have multiple; wrong item selection (231 cases) from complex product catalogs; wrong replacement item (221 cases); wrong payment method (118 cases); other argument mismatches (162 cases); and wrong reason (12 cases). These errors frequently co-occur with MISSING_LOOKUP — agents skip get_order_details calls and then operate on incorrect entity IDs.

NO_CONFIRMATION (626 cases) is the most frequent protocol violation: the agent executes irreversible actions without prompting the user to confirm, which poses a serious operational risk.

5. Key Findings and Insights

Finding 1: Error profiles shift fundamentally with model scale.

4B is dominated by WRONG_TOOL (53.7%) — it tries but picks the wrong tools. 8B/14B/32B are dominated by PREMATURE_TERMINATION (69.3%/37%/35%) — they generate plausible completion text without executing tools. Scaling up model size trades tool selection errors for execution hallucination errors.

Finding 2: Larger models are more prone to hallucinated completion.

32B AUTH_MISSING (17.9%) exceeds 14B (12.1%), and 32B generates the most confident fabricated confirmation messages. The additional capacity enables more convincing but more dangerous failure modes.

Finding 3: Error cascades are common.

AUTH_MISSING rarely appears alone — 80% of auth-missing cases also show PREMATURE_TERMINATION. The cascade: skip auth → fabricate user data → hallucinate completion. Addressing the root cause could eliminate multiple downstream errors.

Finding 4: Infrastructure failures are significant.

EMPTY_TRAJECTORY (26% of failures) represents server crashes, not agent behavior. This inflates failure counts and must be accounted for when comparing error distributions.

Finding 5: Function Calling shows better consistency but a lower peak.

FC's pass^5 degrades less steeply than Act/ReAct, but its pass^1 is often lower. A structured tool-calling format provides more reliable invocations at the cost of constraining the agent's reasoning flexibility.

Finding 6: Post-action verification is critical.

70% of PREMATURE_TERMINATION cases are hallucinated completions detectable only after the agent responds. This motivates post-action verification in our proposed system, unlike IRMA, which is pre-action only.

6. Individual Contributions

Divyesh Patel

Led infrastructure setup and 32B model benchmarking on ASU Sol with NVIDIA A100 GPUs. Designed split-batch execution strategy, implemented shared HuggingFace cache, developed cleanup scripts for port conflicts, and managed Intel Gaudi HPU migration. Produced all 32B baseline results across both domains. For Phase 2: designed the 5-level error taxonomy, and conducted error analysis of 32B trajectories, including manual sub-pattern identification.

Devesh Valluru

Conducted 4B model (Qwen3-4B-Instruct) benchmarking on the airline domain using Gaudi HPU as the primary compute. Tuned memory utilization parameters and parallelism settings. For Phase 2: contributed to trajectory sampling and 4B error pattern analysis, identified the WRONG_TOOL dominance pattern unique to smaller models.

Wei-An Wang

Executed 14B model (Qwen3-14B-AWQ) benchmarking across both domains using quantized deployment on a single A100. Developed automation tooling (check_result_report.py, merge_results.py). For Phase 2: developed the automated error classification scripts (error_analysis_folder.py, error_sampler.py), generated all error analysis plots, created the trajectory extraction pipeline, authored the quantitative analysis sections, and led the cross-model comparison analysis.

Abhishek Mulasi

Conducted 8B model (Qwen3-8B-Instruct) benchmarking. Managed A100 GPU allocation and attempted Gaudi HPU deployment. For Phase 2: contributed to manual trajectory review and failure pattern documentation, assisted with multi-agent system proposal design, and cause-oriented grouping.

7. References

[1] Yao et al., "τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains." NeurIPS 2024. https://github.com/sierra-research/tau-bench

[2] Wang et al., "IRMA: Input-Reformulation Multi-Agent for τ-Bench." arXiv:2508.20931, 2025.

[3] Yao et al., "ReAct: Synergizing Reasoning and Acting in Language Models." ICLR 2023.

[4] Qwen Team, "Qwen3 Technical Report." 2025.